



ACQLERATE

MODULE 2

Defense Finance & Budgeting

Master the PPBE cycle, appropriations law, EVM, and financial management fundamentals that every PM must know.

10 Lessons · ~229 Minutes · Lesson Book Edition

Course Lessons

01	The DoD Acquisition System Overview	12 min
02	ACAT Levels: How the DoD Categorizes Your Program	18 min
03	The PPBE Cycle: Planning Your Program's Money	15 min
04	DoD Appropriations by Service: Color of Money	20 min
05	Cost Estimating & Independent Cost Estimates	24 min
06	Earned Value Management (EVM)	18 min
07	EAC Deep Dive: Forecasting Final Program Cost	22 min
08	Contractor Cost Structure: Wrap Rates, Overhead & G&A	25 min
09	DCAA & DCMA: Your Two Most Important Oversight Partners	20 min
10	Contract Finance Intelligence: CPAF, Burn Rate & Award Fee	55 min

DEFENSE FINANCE & BUDGETING · LESSON 1

The DoD Acquisition System Overview

12 min

Understand the structure, purpose, and key regulations governing DoD acquisitions.

Understanding the Structure, Purpose, and Key Regulations

Defense finance and budgeting sits on top of a regulatory structure every acquisition professional needs to know cold. These are the documents and rules that govern how programs are funded, categorized, and overseen, dense reading that rewards patience and a strong cup of coffee.

- **Title 10 U.S.C.:** Statutory authority for defense acquisitions and armed forces
- **FAR (Federal Acquisition Regulation):** Codified in 48 CFR, governs all federal procurement
- **DFARS (Defense Federal Acquisition Regulation Supplement):** adds 200+ pages of DoD-specific rules on top of the FAR you just finished reading (or “war FARs” once the letterhead catches up)
- **DoDI 5000.02:** The “bible” for major system acquisitions, defines program phases
- **DoDI 5000.74:** Governs Defense Acquisition of Services
- **DoDI 5000.75:** Governs Business Systems acquisitions

ACAT Categories at a Glance

CATEGORY	THRESHOLD	OVERSIGHT LEVEL
ACAT I	RDT&E > \$480M or Procurement > \$2.79B	Milestone Decision Authority: USD(A&S)
ACAT IA	IT: > \$300M total	MDA: ASD(NII) or CIO
ACAT II	RDT&E > \$185M or Procurement > \$835M	MDA: Component (Secretary level)
ACAT III	Below ACAT II thresholds	MDA: Program Executive Officer

The Six Acquisition Pathways

PATHWAY	BEST FOR	KEY CHARACTERISTIC	TYPICAL TIMELINE
Major Capability Acquisition (MCA)	Large, complex systems (ACAT I/II)	Full DoDI 5000 oversight; Milestones A/B/C; most rigorous	10-20+ years
Middle Tier (MTA) - Rapid Prototyping	Fielding prototype within 5 years	No Milestone A; accelerated; used for emerging tech (OTA-friendly)	2-5 years
Middle Tier (MTA) - Rapid Fielding	Fielding proven capability within 6 months-5 years	Leverages existing technology; limited production; no new development	6 months - 5 years
Software Acquisition	Software-intensive programs	Agile delivery; no traditional milestones; continuous iteration	Ongoing sprints
Urgent Capability Acquisition	Validated urgent operational need (UON)	Bypasses normal process; less than 2 years to field	< 2 years
Defense Business Systems	Financial/HR/ERP IT systems	Business case required; milestone reviews adjusted for IT	Varies

Key Terms

FAR

Federal Acquisition Regulation: the primary regulation governing federal procurement.

DFARS

Defense Federal Acquisition Regulation Supplement: DoD-specific additions to FAR.

DoDI 5000.02

The primary DoD instruction governing the acquisition of major defense systems.

JCIDS

Joint Capabilities Integration and Development System: the requirements generation process.

PPBE

Planning, Programming, Budgeting, and Execution: DoD's resource allocation process.

DEFENSE FINANCE & BUDGETING · LESSON 2

ACAT Levels: How the DoD Categorizes Your Program

18 min

Master ACAT levels, milestone decision authority, tailoring, and how program categorization drives oversight, reporting, and your day-to-day responsibilities as a PM.

Why ACAT Level Determines Your Entire Program Experience

The first question asked about any new DoD acquisition is: what ACAT is this program? The answer determines who approves your milestones, how many oversight reviews you will conduct, whether you need an independent cost estimate, whether you must report to Congress, and how much staff support you will have (spoiler: never as much as you actually want). An ACAT I program like the F-35 involves USD(A&S) as the MDA, mandatory SAR reports to Congress, CAPE independent cost estimates, and DoD-level reviews at every milestone. An ACAT III program may be approved by a PEO with minimal external oversight. Understanding ACAT levels is the starting point for understanding any DoD program's regulatory environment.

ACAT Levels: Thresholds, MDA, and Key Requirements

ACAT	COST THRESHOLD (2024)	MDA	KEY REQUIREMENTS	REPORTING
ACAT ID	RDT&E > \$480M or Procurement > \$2.79B; OR designated by USD(A&S)	USD(A&S) or designated SAE	Full DoDI 5000 compliance; ICE required; full documentation	SAR to Congress; unit cost reporting; CAPE ICE
ACAT IC	Same thresholds as ACAT ID	Component Acquisition Executive (CAE) / SAE	Same as ACAT ID minus OSD-level milestone approval	SAR; CAPE ICE optional but often requested
ACAT II	RDT&E > \$185M or Procurement > \$835M	CAE (ASA(ALT), ASN(RDA), SAF/AQ)	Component-level ICE; DoDI 5000 compliance with tailoring	Component-level reporting; no mandatory SAR
ACAT III	Below ACAT I/II; designated by CAE	PEO or designated official	Significant tailoring available; streamlined documentation	Program office reporting only
ACAT IV	Non-major services/ acquisitions	Head of contracting activity	Minimum oversight; acquisition plan required	Internal only

The Adaptive Acquisition Framework: Six Pathways

DoDI 5000.02 (updated in 2020) introduced the Adaptive Acquisition Framework (AAF), replacing the previous “one-size-fits-all” linear acquisition model with six distinct pathways. Each pathway is tailored to a specific type of acquisition: (1) Major Capability Acquisition (MCA), the traditional ACAT process for large, complex defense systems; (2) Middle Tier of Acquisition (MTA), Rapid Prototyping (2 years) or Rapid Fielding (5 years); (3) Software Acquisition, for software-intensive systems using Agile/DevSecOps; (4) Defense Business Systems (DBS), for ERP and business IT systems; (5) Acquisition of Services, for services contracts managed under the service acquisition framework; (6) Urgent Capability Acquisition, for combat-emergent needs addressable within 2 years.

One of the six is literally named Defense Business Systems, so yes, that pun is sitting right there again. We will not dwell on it twice in one book series.

AAF Pathway Selection Guide

PATHWAY	BEST FOR	TIMELINE	KEY ADVANTAGE
Major Capability Acquisition (MCA)	Large, complex systems requiring full R&D lifecycle (aircraft, ships, satellites)	10-20+ years	Full oversight and documentation: appropriate for highest-risk, highest-cost programs
Middle Tier: Rapid Prototyping	Technology demonstrations; filling known capability gaps quickly	≤ 2 years to field	Skip traditional milestones; prototype and learn fast
Middle Tier: Rapid Fielding	Mature technology ready to field with minor modification	≤ 5 years to field	Streamlined acquisition; use existing systems/platforms
Software Acquisition	Software-intensive systems; cloud apps; command and control IT	Continuous delivery cycles	Agile/DevSecOps; iterative delivery; avoid “big bang” software programs
Defense Business Systems	ERP, finance, HR, logistics IT systems	Per commercial release cycle	Use commercial software best practices; avoid military customization
Urgent Capability Acquisition	Combat-emergent gaps; urgent warfighter needs	≤ 2 years	Highest streamlining; near-direct fielding from combatant command request

ACAT Determination Is Not Static: Programs Get Re-Categorized

Programs can be re-categorized upward if cost growth triggers higher ACAT thresholds, which means more oversight, more reporting, and a new MDA. An ACAT II program that grows to ACAT I cost levels gets re-designated, and the PM suddenly must comply with full ACAT I requirements retroactively. This is another reason aggressive cost management is not just financial discipline: it is also a program management imperative to avoid triggering additional oversight burdens.

Selected Acquisition Report (SAR): The ACAT I Report to Congress

For ACAT I programs (MDAPs), the PM must produce a Selected Acquisition Report (SAR) annually, or whenever a Nunn-McCurdy breach or other significant change occurs. The SAR is submitted to Congress and covers: current program description, total program cost (including all variants), cost and schedule changes from original baseline, current schedule, and performance status. The SAR is a legal disclosure document: PMs must ensure all data is accurate. False or misleading SARs can trigger Congressional investigations. For an experienced PM, the SAR is also a professional document that reflects directly on your program's credibility and your ability to manage to a baseline.

Tailoring: Use It, Don't Abuse It

Tailoring under DoDI 5000.02 allows PMs to adjust documentation, review, and reporting requirements based on program risk, complexity, and pathway. A small ACAT III program does not need a 200-page Acquisition Strategy or a full EVMS: tailoring allows appropriate scaling. However, tailoring is a two-way street: the MDA can also impose additional requirements if the program needs more oversight. The best tailoring approach is to clearly justify every decision to omit or reduce a requirement: not simply to skip paperwork but to demonstrate the program has equivalent risk management in place.

Managing Oversight Bodies as a Mid-Career PM

ACAT level determines not just who approves your milestones, but who scrutinizes your program continuously. An ACAT I program will have OSD Cost Assessment and Program Evaluation (CAPE) producing independent cost estimates, DOT&E reviewing your test plans, and OUSD(R&E) evaluating technology maturity. Understanding what each oversight body cares about (and feeding them the data they need proactively) is a mid-career PM survival skill. CAPE cares about cost realism. DOT&E cares about whether testing is operationally realistic. OUSD(R&E) cares about TRL. Brief them early, brief them often, and never surprise them in a milestone review (a DAB is not the place for plot twists). Surprises at DAB reviews kill programs.

Tailoring Acquisition: What Senior PMs Actually Do at Milestone B

The DoDI 5000 framework provides the default acquisition structure, but senior PMs know that almost everything can be tailored. At Milestone B, the MDA approves your Acquisition Program Baseline (APB), your contract strategy, and your tailoring decisions. Smart tailoring means: selecting the minimum oversight that still provides adequate risk visibility, adjusting milestone criteria to match your specific technology maturity, and requesting waivers for documentation requirements that add process burden without insight. The danger is tailoring away safeguards that exist for good reasons. ACAT I programs that waive IBR requirements or skip SEPs have consistently worse cost outcomes. Know which gates exist to protect the program, and which ones are just bureaucratic inertia.

Key Terms

ACAT

Acquisition Category: a classification system that determines the level of oversight, reporting, and milestone decision authority for a DoD acquisition program. Higher ACAT = more oversight.

ACAT I

Major Defense Acquisition Program: total cost > \$480M (RDT&E) or \$2.79B (procurement). MDA is USD(A&S) or a designated Service Acquisition Executive (SAE). Highest oversight.

ACAT II

Major System: total cost > \$185M (RDT&E) or \$835M (procurement). MDA is the DoD Component Acquisition Executive (CAE).

ACAT III

Below ACAT I/II thresholds. MDA is designated by the CAE, typically a Program Executive Officer (PEO) or head of contracting activity.

ACAT IV

Services and non-major acquisitions managed below ACAT III threshold. Component manages with minimal centralized oversight.

MDA

Milestone Decision Authority: the individual with authority to approve milestones, authorize program entry into lifecycle phases, and certify programs.

SAE

Service Acquisition Executive: the senior official responsible for acquisition programs within an Armed Service.

Tailoring

The process of adjusting DoDI 5000.02 requirements to match the program's complexity, acquisition pathway, risk, and urgency.

AAF

Adaptive Acquisition Framework: DoDI 5000.02's six acquisition pathways, allowing programs to select the pathway best suited to their type of acquisition.

MDAP

Major Defense Acquisition Program (yes, there is that word again; if the rename ever finalizes, someone in a Pentagon hallway is going to have to explain “MWAP” with a straight face): an ACAT I program. Subject to full statutory reporting to Congress (SAR, unit cost reporting) and CAPE independent cost estimates at each milestone.

Knowledge Check

An ACAT I program (MDAP) requires which of the following that ACAT III programs do not?

A. A written acquisition strategy

✓ **B. A Selected Acquisition Report (SAR) submitted to Congress and an Independent Cost Estimate at each milestone**

C. A Performance Work Statement

D. A contract administration plan

WHY

ACAT I programs are MDAPs subject to statutory reporting requirements including annual SARs to Congress, CAPE independent cost estimates at each milestone, and Nunn-McCurdy unit cost reporting.

The current ACAT I threshold for Research, Development, Test and Evaluation (RDT&E) is approximately:

A. \$100M

B. \$185M

✓ **C. \$480M**

D. \$2.79B

WHY

Programs with RDT&E costs exceeding approximately \$480M (or procurement costs exceeding \$2.79B) are classified ACAT I.

Under the Adaptive Acquisition Framework, the “Middle Tier: Rapid Prototyping” pathway is designed for programs that can:

A. Take 20 years to deliver a full production system

✓ **B. Field a prototype or demonstration within 2 years, leveraging existing technology**

C. Deliver software via continuous Agile sprints

D. Acquire commercial off-the-shelf services directly

WHY

Rapid Prototyping (AAF Middle Tier) allows programs to field a prototype or proof of concept within 2 years, bypassing traditional Milestone A-B-C review requirements.

If an ACAT II program's costs grow and exceed ACAT I thresholds, what happens?

A. The program is automatically cancelled

✓ **B. The program is re-designated as ACAT I, triggering additional oversight requirements, SAR reporting, and a new MDA**

C. The program continues under ACAT II oversight regardless of cost growth

D. The PM must request a waiver from Congress

WHY

ACAT categorization follows cost: if a program grows beyond the ACAT II threshold into ACAT I territory, it must be re-designated ACAT I, bringing SAR reporting, CAPE ICE, and a new MDA.

Tailoring under DoDI 5000.02 allows a PM to:

A. Eliminate all oversight requirements for expedited programs

✓ **B. Adjust documentation, review, and reporting requirements based on program risk and pathway: with MDA approval**

C. Override Congressional reporting requirements for classified programs

D. Skip the Acquisition Program Baseline for rapid fielding programs

WHY

Tailoring requires MDA approval and must be documented in the Acquisition Strategy. Statutory requirements (SAR, Nunn-McCurdy, ICE for ACAT I) cannot be tailored away.

The Milestone Decision Authority (MDA) for an ACAT ID program is:

A. The Program Manager

B. The Program Executive Officer (PEO)

✓ **C. USD(A&S) or designated Service Acquisition Executive**

D. The Secretary of Defense

WHY

ACAT ID MDAs are USD(A&S) or a designated Service Acquisition Executive (SAE): the most senior acquisition officials in their respective Service.

An Acquisition Program Baseline (APB) documents which three key program parameters?

A. Contract type, vendor selection, and delivery location

✓ **B. Cost objectives (threshold and objective), schedule, and performance (key performance parameters)**

C. Funding levels, test events, and contractor team

D. Congressional appropriations, FYDP profile, and POM submission

WHY

The APB establishes the program's "triple constraint" baseline: cost (threshold and objective values), schedule (key milestone dates), and performance (Key Performance Parameters).

The "Software Acquisition" pathway under the AAF is specifically designed for:

A. Purchasing commercial off-the-shelf software licenses

✓ **B. Software-intensive systems using continuous delivery, Agile development, and DevSecOps practices**

C. Programs that use software as a minor component of a hardware-dominant system

D. Business system acquisitions below \$185M

WHY

The Software Acquisition pathway (DoDI 5000.87) enables Agile/DevSecOps continuous delivery for software-intensive programs: replacing the traditional plan-everything-upfront model.

A Selected Acquisition Report (SAR) is submitted to Congress:

A. Monthly during active program phases

✓ **B. Annually, and additionally whenever a Nunn-McCurdy breach or significant baseline change occurs**

C. Only when a program is cancelled

D. Only at each major milestone decision

WHY

SARs are annual reports to Congress covering ACAT I program cost, schedule, and performance status. Special SARs are required when a Nunn-McCurdy breach occurs.

What distinguishes an “Urgent Capability Acquisition” pathway program from a standard ACAT program?

A. It requires more oversight due to its urgency

✓ B. It allows fielding a capability within 2 years through streamlined approvals, typically driven by combatant command urgent needs

C. It is only available to classified programs

D. It eliminates all contractor competition requirements

WHY

The Urgent Capability Acquisition pathway enables rapid response to urgent warfighter needs (typically within 2 years) using streamlined milestone approvals and expedited contracting.

DEFENSE FINANCE & BUDGETING · LESSON 3

The PPBE Cycle: Planning Your Program's Money

15 min

Understand how DoD programs are funded through the Planning, Programming, Budgeting, and Execution process.

Understand How DoD Programs Are Funded

The PPBE (Planning, Programming, Budgeting, and Execution) process is how DoD programs get and keep their money, year over year, on a timeline generous enough that the requirement can change twice before the budget shows up.

PHASE	WHEN	KEY ACTIVITY	KEY PLAYERS
Planning	Year -2 (Spring)	DPG issued; Services identify capability gaps and priorities	OSD, Joint Chiefs, Service Secretaries
Programming	Year -2 (Summer)	Services submit POMs; OSD issues PBDs	Service HQ, PEOs, Program Offices
Budgeting	Year -1 (Fall/Winter)	Budget Estimate Submissions; OMB passback	Service Comptrollers, OMB, OSD Comptroller
Execution	Current Year	Congress appropriates; PMs execute approved funding	Congress, PMs, Budget Officers

- **RDT&E (Research, Development, Test & Evaluation):** 3-year availability; used to develop and test new systems
- **Procurement:** 3-year availability; used to buy production-ready systems
- **O&M (Operations & Maintenance):** 1-year availability; day-to-day operating costs and services, gone at midnight on September 30 whether you spent it or not
- **MILCON (Military Construction):** 5-year availability; construction of facilities
- **MILPERS (Military Personnel):** 1-year; pay and allowances for uniformed personnel

POM Execution: Managing Your Program Through the Budget Cycle

Mid-career PMs live in two worlds simultaneously: executing the current year's approved budget and defending next year's POM submission. Your POM submission must be defensible with cost data, supported by your program's APB, and internally consistent with your contract structure. The most common POM mistake: projecting out-year costs that don't match your contract's performance schedule, or requesting procurement funding before RDT&E work is complete. Budget analysts at the Component and OSD level will catch these inconsistencies. Build your POM from the contract up: not from the budget down. Cross-walk your WBS to your budget line items. When the two don't align, you get reprogramming problems in execution.

Congressional Marks and Unfunded Priorities: The Hidden Budget Game

The President's Budget Request is a starting position, not an ending position. Congress marks up every defense bill: adding earmarks, cutting programs, fencing funds pending reports. Senior PMs must track their program's budget through the NDAA markup process: House HASC mark, Senate SASC mark, conference report. Fenced funding (congressionally restricted) requires notifications or reports before obligation: failing to track these leads to Anti-Deficiency Act violations. Unfunded Priorities Lists (UPLs) are the Service Chiefs' mechanism for asking Congress to add money above the budget request. If your program is on a UPL, it creates both opportunity (more money) and risk (congressional attention). Know where your program sits in the congressional priority stack: your program's survival may depend on it.

Key Terms

PPBE

Planning, Programming, Budgeting, and Execution: DoD's annual resource management process.

POM

Program Objectives Memorandum: the Services' funding proposals submitted to OSD.

FYDP

Future Years Defense Program: 5-year funding baseline for all DoD programs.

PBD

Program Budget Decision: OSD decisions on Service POMs that shape the President's Budget.

NDAA

National Defense Authorization Act: annual legislation authorizing DoD activities and funding.

DEFENSE FINANCE & BUDGETING · LESSON 4

DoD Appropriations by Service: Color of Money

20 min

Master the specific appropriation accounts used across the Army, Navy, Air Force, and Marine Corps, and why using the wrong “color of money” is a federal violation.

Appropriation Types at a Glance

APPROPRIATION	WHAT IT FUNDS	PERIOD OF AVAILABILITY	KEY RULE
O&M	Operations, training, maintenance, most services contracts under \$350K threshold	1 Year	Cannot fund capital equipment > \$250K (investment threshold) or construction > \$750K
RDT&E	Research, development, prototypes, testing: from basic research through system development	2 Years	Cannot fund production quantities; use ends at Milestone C
Procurement	Production units, end-items: aircraft, ships, missiles, vehicles, major components	3 Years	System must be past Milestone C / production-ready
MILCON	Permanent construction of buildings, infrastructure, and facilities	5 Years	Threshold \$1.5M+; below threshold uses O&M (minor construction)
MILPERS	Military pay, allowances, bonuses, PCS moves	1 Year	Cannot fund civilian personnel or contractor costs

Service Appropriation Accounts

SERVICE / APPROPRIATION	TAS	WHAT IT FUNDS
Air Force O&M	57-3400	AF operations, training, depot maintenance, most AF services
Air Force RDT&E	57-3600	AF research, development, testing: from basic research to EMD
Air Force Aircraft Procurement	57-3010	Procurement of AF aircraft (F-35A, C-130, KC-46, etc.)
Air Force Missile Procurement	57-3020	Procurement of AF missiles (JASSM, AMRAAM, ICBM sustainment production)
Air Force Other Procurement AF	57-3080	AF vehicles, C2 systems, comm equipment, training equipment
Army O&M	21-2010 (2400)	Army operations, training, depot maintenance, most Army services
Army RDT&E	21-0400 (2040)	Army research and development
Army Aircraft Procurement	21-2010	Army rotary-wing aircraft (AH-64, UH-60, CH-47)
Army Missile Procurement	21-2020	Army missiles (Patriot, Stinger, Javelin production)
Army Procurement of W&TCV	21-2035	Wheeled & tracked combat vehicles (M1 Abrams, Bradley, Stryker)
Army Other Procurement Army	21-2060	Army C4I, aircraft modifications, soldier systems
Navy O&M (Navy)	17-1453	Navy operations, training, base support
Navy O&M (Marine Corps)	17-1105	USMC operations and training
Navy Aircraft Procurement Navy	17-1506	Navy/USMC aircraft (F/A-18, F-35B/C, E-2D, P-8)
Navy Weapons Procurement Navy	17-1507	Navy weapons (Tomahawk, Standard Missile, torpedoes)
Navy Shipbuilding & Conversion	17-1611	Construction and conversion of Navy vessels (CVN, DDG, SSN)
Navy Other Procurement Navy	17-1804	Navy C4ISR, shore activities, training equipment
Navy Procurement Marine Corps	17-1109	USMC procurement (AAV, LAV, ground equipment)
Defense-Wide O&M	97-0400 (0100)	DISA, DLA, SOCOM, and other defense agencies operations
Defense-Wide RDT&E	97-0400 (0603)	DARPA, MDA, and other defense agency R&D

Managing Color of Money Across a Complex Program

Real programs use multiple appropriation types simultaneously. A typical ACAT II program in EMD might have RDT&E funding the development contract, O&M funding government lab support, and MILCON funding a test facility. Each appropriation has different obligation and expenditure rules, think of it less like a shared household budget and more like separate vending machines that each take exact change in a different currency. The most common mid-career mistake: using RDT&E to fund activities that have crossed into production (violates the appropriation's purpose), or letting O&M funds expire while trying to transfer them to RDT&E (not permitted). Your comptroller is your best friend, but you need to give them enough lead time to structure funding actions properly. Last-minute funding transfers almost always fail or create audit exposure.

Anti-Deficiency Act Violations and How They Happen to Good PMs

The Anti-Deficiency Act (31 U.S.C. § 1341) prohibits obligating funds in excess of or in advance of appropriations. Violations are career-ending. They happen most often through: (1) continuing contract performance after a CR expires and before new appropriations are enacted without a proper Continuing Resolution Authority authorization; (2) using one appropriation to fund activities that belong to another (“purpose” violations); (3) obligating contracts before funds are available in USASpending. None of these start with someone twirling a mustache; they start with someone being three days late updating a spreadsheet. Senior PMs must build a personal funding calendar: when does each appropriation expire? When does the CR end? What is the period of availability for each line item? These are not comptroller problems: they are PM problems. The PM signs the obligation documents.

Key Terms

TAS

Treasury Account Symbol: the unique identifier for each appropriation account (e.g., AF 3400 for Air Force O&M).

O&M

Operations & Maintenance: 1-year appropriation funding day-to-day operations, maintenance, and most services contracts.

RDT&E

Research, Development, Test & Evaluation: 2-year appropriation funding research and development activities.

Procurement

3-year appropriation funding production and purchase of end-items (aircraft, ships, missiles, vehicles).

MILCON

Military Construction: 5-year appropriation funding construction of military facilities.

Bona Fide Need Rule

Funds may only be used to satisfy a legitimate, genuine need that arose during the period of availability.

Purpose Statute

31 U.S.C. § 1301: appropriations may only be applied to the objects for which they were made.

Color of Money

The appropriation type funding a particular contract line: mixing appropriation types on one CLIN is illegal.

Expired Funds

Funds past their period of availability but still available for 5 years to adjust or pay existing obligations.

Canceled Funds

Funds that have completed the 5-year expired period; permanently returned to the Treasury.

DEFENSE FINANCE & BUDGETING · LESSON 5

Cost Estimating & Independent Cost Estimates

24 min

Master cost estimating methodologies, the CAPE independent cost estimate process, and (critically) how contractors price labor through LCATs, pre-priced labor, job codes, and compensation ratios that directly drive your program cost.

Cost Estimating Methodologies

- **Analogous (Top-Down):** Uses historical data from similar programs. Fast but less accurate, the estimating equivalent of guessing a stranger's age. Best for early phase estimates ($\pm 50\%$).
- **Parametric:** Uses Cost Estimating Relationships (CERs): statistical models built from large databases. Moderately accurate. Best for early-mid phase ($\pm 25\%$).
- **Engineering Build-Up (Bottoms-Up):** Detailed task-by-task estimate. Most accurate but requires mature design. Best for milestone B+ ($\pm 10\text{-}15\%$).
- **Hybrid approaches:** Most programs use a combination: parametric for unknowns, bottoms-up for well-defined work.

Sample Labor Category (LCAT) Rate Card

LCAT TITLE	SKILL LEVEL INDICATORS	TYPICAL DIRECT LABOR RATE (\$/HR)	FULLY BURDENED RATE EXAMPLE
Program Manager I	BS + 5 yrs; manages subteams; no independent authority	\$65-85/hr	\$165-215/hr (1.5-2.5x wrap)
Program Manager II	BS + 10 yrs or MS + 8 yrs; manages programs <\$50M	\$90-120/hr	\$225-300/hr
Program Manager III	BS + 15 yrs or MS + 12 yrs; manages programs >\$50M; strategic decisions	\$130-175/hr	\$325-440/hr
Systems Engineer II	BS + 4-7 yrs; requirements analysis, design trade studies	\$70-90/hr	\$175-225/hr
Systems Engineer III	BS + 8-12 yrs or MS + 5 yrs; system architecture, integration lead	\$95-130/hr	\$240-325/hr
Software Engineer II	BS CS/CE + 3-6 yrs; coding, testing, integration	\$75-100/hr	\$190-250/hr
Financial Analyst I	BS Finance/Accounting + 2-4 yrs; budget tracking, reporting	\$55-70/hr	\$138-175/hr
Contracts Specialist II	BS + 5 yrs contracting; proposal prep, mods, compliance	\$65-85/hr	\$163-213/hr

Five Questions to Ask When Reviewing Proposed Labor Rates

1. Does the LCAT mix make sense for the statement of work? Senior engineers should not dominate routine maintenance tasks.
2. Are the proposed direct labor rates consistent with the contractor's disclosed forward pricing rates (FPRAs)? If not, why not?
3. For IDIQ/pre-priced vehicles: are the proposed rates at or below the contract ceiling rates? Any ceiling rate exception requires CO approval.
4. Is the contractor proposing the same LCATs they actually staff? Request key personnel resumes and verify job codes match proposed LCATs.
5. What is the annual escalation rate applied to labor in out-years? Defense labor typically escalates 3-4% annually; proposals using 0% escalation are understating future costs.

Defending Your Cost Estimate in a CAPE Review

The OSD Cost Assessment and Program Evaluation (CAPE) office produces Independent Cost Estimates (ICEs) for major programs at each milestone. When CAPE's ICE is significantly higher than your Program Office Estimate (POE), the MDA must choose which to use for the APB, and MDA almost always uses the higher one (CAPE has never, in recorded history, come in lower than the program office estimate). Mid-career PMs need to understand how CAPE builds its ICEs: parametric models (SEER, PRICE, COCOMO for software) calibrated to historical analogues. To reduce the gap, you must document your differences with CAPE: what makes your program different from the analogues, what risk mitigations justify lower estimates, and what ground rules and assumptions diverge. The goal isn't to beat CAPE: it's to narrow the gap through credible technical justification.

Should-Cost vs. Will-Cost: The Senior PM's Most Powerful Tool

Should-Cost analysis is a government-led deep dive into contractor cost structure to identify efficiencies: not just to project what costs will be, but to drive them down. DoDI 5000.02 requires Should-Cost management on major programs. In practice, should-cost teams examine contractor labor mix (are they using senior engineers where journeymen would do?), overhead allocation (is G&A allocation methodology favorable to the program?), and make-buy decisions. The results feed should-cost targets: negotiating positions for the next contract award. Senior PMs who run aggressive should-cost programs routinely achieve 10-15% cost reductions. The political challenge: your contractor will resist deeply, and your own program office staff (who work with the contractor daily) will resist disrupting the relationship. Push anyway.

Key Terms

ICE

Independent Cost Estimate: an estimate prepared independently of the program office, required at key milestones for ACAT I programs.

CAPE

Cost Assessment and Program Evaluation: OSD office responsible for independent cost analysis of major defense acquisition programs.

Analogous Estimating

Using costs from similar past programs to estimate the new program. Fast but least accurate ($\pm 50\%$). Best for pre-Milestone A.

Parametric Estimating

Using statistical Cost Estimating Relationships (CERs) to relate cost to technical parameters. Moderate accuracy ($\pm 25\%$).

Bottoms-Up Estimating

Estimating each work package or task individually and rolling up to a total. Most accurate ($\pm 10\text{-}15\%$) but requires mature design.

Cost Risk

The uncertainty or variability in a cost estimate; typically quantified at the 80th percentile (P80) for DoD program budgeting.

LCAT

Labor Category: a defined skill level and discipline that establishes what work a person performs and at what rate they are billed.

Pre-Priced Labor

Labor categories with negotiated, fixed billing rates established in the base contract, allowing task orders to be placed without re-pricing labor.

Compensation Ratio (Compa-Ratio)

An employee's actual salary divided by the midpoint of their pay band. Used to assess whether a contractor's labor pricing is fair and competitive.

Job Code

A contractor-internal identifier linking an employee to a specific LCAT and pay band. Audited by DCAA to verify labor charging accuracy.

Direct Labor Rate

The base hourly wage paid to an employee before fringe benefits or overhead are applied.

Fully Burdened Rate

The total cost per labor hour including all fringe benefits, overhead, G&A, and fee. What the government actually pays per hour of contractor labor.

DEFENSE FINANCE & BUDGETING · LESSON 6

Earned Value Management (EVM)

18 min

Learn how to use EVM to measure cost and schedule performance on DoD contracts.

EVM Health Indicators at a Glance

INDICATOR	GREEN	YELLOW	RED
CPI	≥ 0.95	0.90 - 0.94	< 0.90
SPI	≥ 0.95	0.90 - 0.94	< 0.90
CV%	$\leq 5\%$ over	5-10% over	$> 10\%$ over
TCPI	≤ 1.05	1.05 - 1.10	> 1.10

Using EVM Data to Drive Contractor Behavior

EVM data is only valuable if the government PM uses it to hold contractors accountable. Mid-career PMs should establish a monthly rhythm: review IPMR data within 5 days of receipt, convene a PMR within 10 days, and issue formal corrective action requests for threshold breaches within 15 days. The most powerful question you can ask a contractor at a PMR: “Your Format 5 says you'll recover 15 schedule days by end of quarter: what specific tasks will be completed early and by whom?” Vague recovery plans are not plans, they are wishes with a Gantt chart attached. Require specificity: named individuals, milestone dates, budget planned for the recovery effort. Document these commitments in meeting minutes so the contractor can't walk them back.

EVM Compliance Surveillance and EVMS Acceptance: What Senior PMs Must Know

DoDI 5000.02 requires EVMS on cost-type contracts over \$20M and fixed-price over \$50M. But having EVMS is not enough: the system must be compliant with ANSI/EIA-748. DCMA conducts EVMS surveillance and can issue non-compliance findings. A Level III finding (systemic noncompliance) can result in a compliance improvement plan, increased surveillance, and in extreme cases, withholding of fee. Senior PMs must understand their contractor's EVMS compliance status before accepting performance data. An EVMS system that rubber-baselines (retroactively adjusts PMB to reduce variances) produces data that is technically compliant but strategically meaningless. When CPI is suspiciously stable at 1.00 month after month on a complex development contract, question the system: not just the data.

Key Terms

EVM (Earned Value Management)

The system that tells you whether a program is getting the work done it planned to get done, at the cost it planned to spend.

BCWS (PV)

Budgeted Cost of Work Scheduled (Planned Value): what you planned to spend by now.

BCWP (EV)

Budgeted Cost of Work Performed (Earned Value): the budget value of work actually accomplished.

ACWP (AC)

Actual Cost of Work Performed (Actual Cost): what you actually spent.

CPI

Cost Performance Index: efficiency of cost performance (EV/AC).

SPI

Schedule Performance Index: efficiency of schedule performance (EV/PV).

EAC

Estimate at Completion: the projected total cost to complete the program.

DEFENSE FINANCE & BUDGETING · LESSON 7

EAC Deep Dive: Forecasting Final Program Cost

22 min

Master all four EAC methods, understand when to use each, and learn how senior leaders use EAC to make go/no-go program decisions.

Why EAC is the Number That Keeps PMs Up at Night

The Estimate at Completion is the most scrutinized number in any program review. It tells leadership one thing: how much is this program actually going to cost? (Humanity has never been great at answering that question honestly, and defense programs have not broken the streak.) Unlike the BAC (your original plan), the EAC incorporates real performance data: what you have spent and what you have accomplished. Senior leaders, PEOs, and Congress use the EAC to decide whether to continue, restructure, or cancel a program. If your EAC exceeds the BAC by 25%, you are in Nunn-McCurdy territory. If it exceeds it by enough to exceed Available Congressional Budget Authority, you may face a program termination. Getting your EAC right (and updating it honestly every reporting period) is one of the most important things a PM does.

The Golden Rule of EAC

Never manage to the EAC you want; manage to the EAC the data tells you. PMs who “sandbag” EACs to avoid leadership scrutiny are not protecting their programs: they are destroying their credibility. Leaders at the PEO and SAE level have seen thousands of programs. They know when an EAC is unrealistic, and a PM who consistently produces optimistic EACs that then miss is far more damaging than one who delivers accurate, uncomfortable forecasts early.

EAC Method Selection Guide

METHOD	ASSUMPTION	BEST FOR	RED FLAG IF USED WHEN...
BAC / CPI	Efficiency stays same as today	Mature programs ($\geq 20\%$), systemic issues	Program is $< 20\%$ complete; single anomaly caused overrun
AC + (BAC-EV)	Remaining work on original budget	One-time anomaly; clear corrective action	CPI has been < 0.9 for 3+ consecutive periods
AC + (BAC-EV)/CPI	Future at current cost efficiency	Standard alternative to Method 1	SPI is also significantly < 1.0
AC + (BAC-EV)/(CPI $\times$ SPI)	Both cost and schedule problems continue	Programs with dual cost+schedule problems	CPI and SPI are ≥ 1.0 (produces artificially low EAC)

How Senior Leaders Read EAC Trends

Program executives and PEOs do not look at a single EAC number: they look at EAC trends over time. A steadily rising EAC signals a systemic problem that corrective action is not fixing. A “hockey stick” EAC (stable for months, then sharp jump) signals that someone was managing the EAC rather than reporting reality. Leaders expect EAC to be updated every IPMR reporting period and to reflect current risk. If your EAC has not moved in 6 months while the program has cost and schedule variances, expect scrutiny. The EAC should be living data.

EAC as a Leadership Decision Tool

EAC SIGNAL	WHAT LEADERSHIP SEES	LIKELY ACTION
EAC rising gradually, TCPI still achievable	Honest reporting; manageable growth	Monitor; request corrective action plan
EAC static despite CPI < 0.90 for 6+ months	Sandbagged EAC; PM may be protecting program	Independent review; replace EAC with BAC/CPI
EAC jumps sharply in one period (“hockey stick”)	Prior periods were optimistic; crisis acknowledgment	DepSecDef-level review; potential program breach
EAC $> 125\%$ of original baseline	Nunn-McCurdy Critical Breach threshold approaching	Congressional notification; USD(A&S) certification required
TCPI > 1.10 on published EAC	EAC is not credible; PM cannot justify forecast	OSD CAPE independent estimate required

PM Best Practice: Maintain Three EAC Scenarios

The most credible PMs bring three EAC scenarios to every senior review: (1) Best Case, assumes corrective actions succeed; (2) Most Likely, Method 1 or 3 based on current data; (3) Worst Case, Method 4 including schedule delays. Showing the range demonstrates analytical rigor and gives leadership the full picture. The worst thing a PM can say to a PEO is “I didn't see it coming.” Three-scenario EAC reporting eliminates that excuse.

EAC vs. Government EAC vs. Independent Agency EAC

Know the difference: (1) Contractor EAC (what the contractor reports in the IPMR Format 1. (2) Government EAC) what the PM's office independently calculates using the same data. These often differ. (3) AFCAA/Service Cost Center EAC: independent agency estimate. When all three diverge significantly, leadership loses confidence in the program. The PM must reconcile differences and explain them clearly.

EAC Reconciliation: When Government and Contractor Disagree

Every IPMR submission includes the contractor's EAC. The government PM should independently calculate a statistical EAC ($BAC \div CPI$, or $BAC \div (0.2 \times SPI + 0.8 \times CPI)$) and compare it to the contractor's number. A contractor EAC significantly below the statistical EAC is a red flag: the contractor may be sandbagging risk or building an optimistic recovery plan. When the gap exceeds 5-10%, require a written EAC reconciliation: line-by-line explanation of why the contractor's estimate differs from the statistical prediction, including what specific actions justify the optimism. Undocumented optimism in an EAC is a program risk, not a performance metric.

Over Target Baseline (OTB) and Over Target Schedule (OTS): The Nuclear Option

When a program's EVM data has lost informational value because the baseline is so unrealistic that CPI and SV are meaningless, the government can authorize an Over Target Baseline (OTB) reprogramming. This resets the PMB, but does NOT change the contract price or schedule. It is a measurement reset, not a bailout. OTB requires MDA/SSA approval, a new Integrated Baseline Review, and a formal reprogramming plan. An OTS (Over Target Schedule) similarly resets time-phasing without changing the contract completion date. Senior PMs use OTB reluctantly: it signals that earlier oversight failed, creates congressional scrutiny, and is frequently reported in DAES and SARs. But refusing to authorize OTB when the baseline is clearly broken means making decisions based on meaningless data. Know when to pull the trigger.

Key Terms

EAC

Estimate at Completion: the total forecasted cost to complete the entire program, calculated using current performance data.

BAC

Budget at Completion: the total approved budget for all authorized work on the program (the baseline).

ETC

Estimate to Complete: the projected cost of work remaining; $ETC = EAC - AC$.

VAC

Variance at Completion: the projected difference between BAC and EAC; $VAC = BAC - EAC$. Negative means overrun.

TCPI

To-Complete Performance Index: the cost efficiency required on all remaining work to achieve the EAC goal: $TCPI = (BAC - EV) / (EAC - AC)$.

CPI

Cost Performance Index: current cost efficiency: EV / AC . The most reliable single predictor of final EAC.

SPI

Schedule Performance Index: current schedule efficiency: EV / PV . Used in combined EAC formulas.

IPMR Format 5

Problem Analysis Report: the IPMR format where contractors explain EAC variances, root causes, and corrective actions.

Knowledge Check

A program is 35% complete with CPI = 0.82 and BAC = \$150M. Using the most statistically accurate EAC formula, what is the projected final cost?

- A. \$123M
- B. \$150M
- ✓ C. \$182.9M
- D. \$210M

WHY

$EAC = BAC / CPI = \$150M / 0.82 = \$182.9M$. Since the program is >20% complete and the CPI has likely stabilized, the BAC/CPI method is the most statistically reliable predictor. The projected overrun is \$32.9M: a 21.9% cost growth.

When is EAC Method 2 (AC + BAC - EV) most appropriately used?

- A. Any time the program has a negative cost variance
- ✓ B. When a specific one-time event caused the overrun and it is not expected to recur
- C. When the program is less than 10% complete
- D. When the TCPI is greater than 1.10

WHY

Method 2 assumes the remaining work will be completed at the originally budgeted rates: implying current inefficiency was a one-time anomaly. It requires strong justification and documentation. Using it when CPI has been consistently below 0.9 for multiple periods is not defensible.

A program has: BAC = \$100M, EV = \$30M, AC = \$40M, EAC = \$130M. What is the TCPI?

- ✓ A. 0.78
- B. 1.00
- C. 0.78 (remaining work at 78% efficiency)
- D. 1.56

WHY

$TCPI = (BAC - EV) / (EAC - AC) = (\$100M - \$30M) / (\$130M - \$40M) = \$70M / \$90M = 0.778$. This means the program only needs 77.8% cost efficiency on remaining work to achieve its EAC: actually achievable since current CPI is $30/40 = 0.75$, and TCPI is slightly higher but comparable.

A TCPI of 1.25 on a published contractor EAC should raise what concern?

A. The EAC is too conservative

✓ **B. The EAC is not credible: it requires 25% better efficiency than current performance, which is historically unprecedented**

C. The contractor needs to increase staff

D. The TCPI calculation uses the wrong formula

WHY

A TCPI above 1.10 is widely considered unrealistic. A TCPI of 1.25 means the contractor must perform 25% more efficiently on all remaining work than they have to date: an extreme assumption that is almost never realized. This signals the EAC is understated and requires independent review.

VAC (Variance at Completion) = BAC - EAC. If BAC = \$200M and EAC = \$235M, what does the VAC tell you?

A. The program will save \$35M

✓ **B. The program is projected to overrun by \$35M at completion**

C. The program has already overspent by \$35M

D. The schedule will slip by \$35M worth of work

WHY

$VAC = \$200M - \$235M = -\$35M$. A negative VAC is a projected cost overrun: the EAC exceeds the BAC. This does not mean money has already been spent; it is a forecast of where the program will end up. The PM must identify corrective actions to reduce this projected overrun.

Which EAC formula incorporates BOTH cost and schedule inefficiency into the remaining work forecast?

A. $EAC = BAC / CPI$

B. $EAC = AC + (BAC - EV)$

C. $EAC = AC + [(BAC - EV) / CPI]$

✓ **D. $EAC = AC + [(BAC - EV) / (CPI \times SPI)]$**

WHY

Method 4 divides remaining work ($BAC - EV$) by the combined $CPI \times SPI$ factor, making it the most conservative estimate. It reflects the reality that a program with both cost overruns and schedule delays will face compounding inefficiency on future work: both spending more per unit of work AND doing that work more slowly.

An EAC that has not changed across 8 consecutive IPMR reporting periods despite a CPI of 0.78 throughout is MOST likely to signal:

A. The program is being managed very effectively

✓ **B. The contractor is sandbagging: managing the EAC number rather than reporting reality**

C. The CPI formula is being applied incorrectly

D. The contract type has been changed to FFP

WHY

When CPI consistently falls below 1.0 but EAC stays flat, the EAC is not reflecting actual performance trends: it is being “managed.” This is a red flag. Leaders should request an independent EAC using BAC/CPI and compare it to the contractor’s reported EAC. A large discrepancy warrants a program review or DCMA surveillance action.

ETC (Estimate to Complete) is defined as:

A. The total projected cost at program completion

B. The cost of work performed to date

✓ **C. The cost of all remaining work: EAC minus AC**

D. The budget set aside for management reserve

WHY

$ETC = EAC - AC$. It is the forecasted cost of all work remaining to be done. ETC is distinct from EAC (total forecast) and AC (actual spend to date). When briefing leadership: “We have spent \$50M (AC), we project a final cost of \$130M (EAC), which means we have \$80M (ETC) of work remaining.”

At what program completion percentage does Christensen's research show the CPI becomes a reliable predictor of final cost efficiency?

A. 5%

B. 10%

✓ **C. 20%**

D. 50%

WHY

Christensen's 1993 research (the most cited study in EVM literature) established that the CPI at 20% completion is highly predictive of final CPI, and that CPI almost never improves significantly after that point. This is why early EVM data is so valuable: a CPI of 0.80 at 20% should be treated as a strong signal the program will finish at approximately that efficiency, not recover to 1.0.

Which IPMR format is specifically designed for contractor explanation of significant cost and schedule variances?

A. Format 1 (WBS-based cost)

B. Format 3 (Baseline)

✓ **C. Format 5 (Problem Analysis)**

D. Format 6 (Milestone/IMS)

WHY

IPMR Format 5 is the Problem Analysis Report: it requires detailed variance analysis, root cause identification, and corrective action plans for variances exceeding government-specified thresholds (typically $\pm \$X$ million or $\pm Y\%$ variance). This is the format PMs read most carefully at monthly program reviews.

DEFENSE FINANCE & BUDGETING · LESSON 8

Contractor Cost Structure: Wrap Rates, Overhead & G&A

25 min

Understand how defense contractors build their prices: wrap rates, fringe, overhead, G&A, fee/profit, and how PMs use this knowledge to evaluate proposals and control costs.

Why PMs Must Understand Contractor Cost Structure

Most PMs focus entirely on total contract price, but the price is built from dozens of individual cost elements that a contractor controls. Understanding how a contractor builds their price gives you critical leverage: you can identify inflated overhead, challenge unrealistic rates, spot rate trends that predict cost growth, and negotiate better deals. More importantly, when DCAA audits a contractor's accounting system and finds non-compliant cost pools or unsupported rates, your contract is at risk. Knowing the structure means you can ask the right questions before signing.

Direct Costs: What the Contractor Directly Charges Your Contract

Direct costs are the foundation of any cost estimate. Direct Labor: the hours your program pays for directly, billed at the contractor's direct labor rate for each labor category (e.g., Systems Engineer Level III: \$75/hour direct). Direct Materials: hardware, software licenses, test equipment, and supplies that can be traced to your contract. Other Direct Costs (ODCs): travel, subcontractor costs, consultant fees, special test equipment. Direct costs are identifiable, auditable, and must be allocable to your contract under CAS and FAR Part 31.

Overhead vs. G&A: The Critical Distinction

Overhead and G&A are both indirect costs, but they are structured very differently. Overhead is typically division or department-level: it covers the cost of running a specific business unit (facilities for that division, department manager salaries, indirect engineers, IT servers for that group). A company with three divisions will have three separate overhead pools, each with a different rate. G&A is company-wide: it covers the CEO's salary, corporate HR, corporate finance, corporate legal, and business development costs that benefit the entire enterprise (your contract is, in a very real sense, chipping in for the corporate holiday party). G&A is applied to Total Cost Input (direct + overhead) as the final layer before fee. Understanding this two-layer structure is critical for evaluating proposals.

Forward Pricing Rate Agreements (FPRAs): What They Are and Why They Matter

An FPRA is a negotiated agreement between a large contractor and DCAA/ACO that establishes the indirect cost rates to be used for pricing future proposals for a set period (typically 1-2 years). Once an FPRA is in place, contractors must use those rates in all proposals: they cannot negotiate rates individually on each contract. For PMs, FPRAs mean two things: (1) you know what rates to expect from major contractors: they are public record in FPRA letters; (2) if a contractor's actual rates come in higher than their FPRA rates, the government may be owed money via an equitable adjustment.

Cost Accounting Standards (CAS): What PMs Need to Know

STANDARD	SUBJECT	PM RELEVANCE
CAS 401	Consistency in estimating, accumulating, and reporting costs	Contractor must estimate costs the same way they accumulate them: prevents gaming
CAS 402	Consistency in allocating costs: direct vs. indirect	Costs cannot be both direct on one contract and indirect on another
CAS 403	Allocation of home office expenses	Controls how corporate G&A is allocated across business units
CAS 404	Capitalization of tangible assets	Defines what gets capitalized vs. expensed: prevents inflating direct costs
CAS 410	Allocation of business unit G&A	Governs the G&A rate structure and base
CAS 418	Allocation of direct and indirect costs	Most frequently cited in DCAA audits: cost pools must be consistent

Reading a Contractor's Incurred Cost Submission (ICS)

Every year, contractors on cost-reimbursable contracts must submit an Incurred Cost Submission (ICS, also called an Incurred Cost Proposal or ICP) showing their actual indirect cost rates for the year. DCAA audits these submissions to verify that final rates match what was billed to contracts. If actual overhead ends up lower than the rates billed to your contract, the contractor owes the government a refund. If actual rates are higher, the government may owe the contractor more. As a PM, you should know your major contractors' ICS filing status: delayed ICS submissions (often years late) are a compliance red flag.

Unallowable Costs: The FAR Part 31 Trap

FAR Part 31 defines which costs are allowable on government contracts. Certain costs are expressly unallowable: meaning the contractor CANNOT bill them to government contracts, even if they are legitimately incurred. Key unallowable costs: entertainment and gifts, lobbying and political contributions, alcoholic beverages, certain advertising costs, costs of organizing or reorganizing the contractor's business, excessive executive compensation above the statutory cap (currently ~\$600K/year for the 5 highest-paid executives), fines and penalties. If DCAA discovers unallowable costs in an incurred cost audit, the contractor must reimburse the government. PMs should flag unusually high G&A rates, which may indicate unallowable costs improperly bundled into cost pools.

Auditing Contractor Indirect Rates During Performance

Mid-career PMs on cost-type contracts must understand that the rates billed on invoices are provisional: final rates are not established until DCAA audits and negotiates final billing rates, typically 2-3 years after the accounting period closes. This creates a reconciliation risk: if final rates are higher than provisional rates, the contractor gets a retroactive billing increase. Track your contractor's provisional vs. forward pricing rate agreements (FPRAs). When provisional rates diverge significantly from actuals, push for an FPRA update. PMs who ignore this find themselves in a final rate reconciliation dispute 3 years after contract closeout: when the contractor is trying to bill \$2M in additional indirect costs that weren't budgeted.

Unallowable Costs, Direct Charging, and the DCAA Audit Trail

FAR Part 31 defines cost allowability: what contractors can bill to the government. Unallowable costs (entertainment, lobbying, certain IR&D that doesn't qualify, executive compensation above the cap) must be segregated and excluded from billings. DCAA's job is to verify this segregation. What senior PMs need to know: if DCAA finds unallowable costs billed to your contract, the contractor must repay them with interest, and the PM has an obligation to pursue collection. The more dangerous scenario is “direct charging” problems: a contractor billing costs directly to your contract that should be in indirect pools (inflating the direct cost base artificially). This is fraud when intentional. Look for anomalous direct labor charges: senior executives billed at 100% direct to a single contract, or subcontractor pass-through markups that exceed the prime's normal margins.

Key Terms

Direct Cost

A cost that can be specifically identified with a single final cost objective (a contract). Examples: direct labor hours, direct materials, travel directly supporting a contract.

Indirect Cost

A cost that cannot be directly attributed to a single contract and must be allocated across multiple cost objectives. Examples: facility rent, HR, finance, IT infrastructure.

Fringe Benefits Rate

The rate applied to direct labor to cover employee benefits: health insurance, retirement, FICA, paid leave. Typically 25-35% of direct labor.

Overhead Rate

Indirect costs allocated to direct labor or direct costs within a specific organizational unit (division, department). Covers facilities, supervision, equipment, and indirect labor.

G&A Rate

General and Administrative rate: company-wide indirect costs covering executive management, corporate HR, legal, finance, and business development. Applied to total cost input.

Wrap Rate

The total billing rate for one hour of labor, combining direct labor + fringe + overhead + G&A + fee. The all-in cost of one hour of contractor work.

Fee / Profit

The contractor's profit on a contract. On cost-type contracts, fee is negotiated separately; on fixed-price contracts, profit is embedded in the price. DFARS limits fee rates by contract type.

Cost Pool

A grouping of indirect costs that are accumulated and then allocated using a common allocation base (e.g., a facilities cost pool allocated based on square footage).

Allocation Base

The measure used to distribute indirect costs to contracts (e.g., direct labor hours, direct labor dollars, total cost input).

CAS

Cost Accounting Standards: 19 standards (48 CFR 9900) governing how defense contractors accumulate, measure, and allocate costs. Required for contracts over \$2M.

CASB

Cost Accounting Standards Board: the federal board that promulgates CAS. Contractors must disclose their accounting practices in a Disclosure Statement (CASB DS-1).

Forward Pricing Rate Agreement

FPRA: a negotiated agreement between the contractor and the government on indirect cost rates for use in forward pricing of proposals. Eliminates rate negotiation on every proposal.

Knowledge Check

A contractor's Systems Engineer earns \$70/hr direct labor. Fringe = 33%, Overhead = 55% of DLR, G&A = 13% of total cost input, Fee = 9%. What is the approximate wrap rate?

A. \$70/hr

B. \$148/hr

✓ C. \$168/hr

D. \$192/hr

WHY

Step by step: DLR = \$70. + Fringe (33%): $70 \times 0.33 = \$23.10 \rightarrow \93.10 . + Overhead (55% of DLR): $70 \times 0.55 = \$38.50 \rightarrow \131.60 . + G&A (13% of \$131.60): $131.60 \times 0.13 = \$17.11 \rightarrow \148.71 . + Fee (9%): $148.71 \times 0.09 = \$13.38 \rightarrow \$162.09 \approx \$162/\text{hr}$. Closest to \$168/hr at these rounding conventions. The multiplier is approximately 2.4× DLR: typical for a mid-size contractor.

Which cost element covers company-wide expenses like executive management, corporate legal, and enterprise IT: applied as the final indirect rate layer before fee?

A. Fringe Benefits

B. Overhead

✓ C. G&A (General and Administrative)

D. Management Reserve

WHY

G&A covers company-wide administrative costs that benefit the entire enterprise: not just a single division. It is applied to Total Cost Input (all direct + overhead costs) as the final indirect cost layer before fee. Overhead is division/department-level; G&A is corporate-level.

Under FAR 15.404-4, the maximum fee on a Cost-Plus-Fixed-Fee (CPFF) completion contract is:

- A. 5% of estimated cost
- B. 7% of estimated cost
- ✓ C. 10% of estimated cost
- D. 15% of estimated cost

WHY

CPFF completion contracts are capped at 10% of estimated cost per FAR 15.404-4(c)(4)(i). CPFF term contracts are capped at 7%. These caps exist because on cost-reimbursable contracts, there is no upper limit on cost: unlimited fee on unlimited cost would remove all incentive for cost control.

An FPRA (Forward Pricing Rate Agreement) benefits the government primarily by:

- A. Allowing contractors to charge any rate they choose per contract
- ✓ B. Establishing negotiated indirect rates applicable to all future proposals, eliminating per-contract rate negotiation
- C. Capping profit at 10% on all contracts
- D. Requiring DCAA audit of every individual proposal

WHY

An FPRA establishes pre-negotiated indirect rates (overhead, G&A, fringe) that the contractor must use in all proposals for the agreement period. This saves significant negotiation time, provides rate predictability, and allows the government to evaluate proposals against known rates rather than contractor-asserted rates.

Under CAS 402, a contractor discovers it has been classifying a particular engineer's time as "indirect" on commercial contracts but "direct" on government contracts. This is:

- A. Acceptable as long as total costs are the same
- ✓ B. A CAS 402 violation: costs must be treated consistently as direct or indirect across all contracts
- C. Permitted since commercial contracts have different rules
- D. Only a problem if DCAA identifies it

WHY

CAS 402 requires consistent treatment of similar costs: a cost cannot be treated as direct on government contracts and indirect on commercial contracts (or vice versa). This is one of the most frequently cited CAS violations in DCAA audits. Inconsistent treatment allows contractors to maximize government billings by allocating their most expensive costs directly to government contracts.

Which of the following is expressly UNALLOWABLE under FAR Part 31 and cannot be billed to government contracts?

- A. Direct engineer labor on contract tasks
- B. Fringe benefits for employees working on the contract

✓ C. Entertainment costs and gifts

- D. Facilities overhead for space used to perform the contract

WHY

FAR 31.205-14 expressly identifies entertainment costs as unallowable. Other key unallowable costs include: lobbying (31.205-22), alcoholic beverages (31.205-51), advertising costs not related to recruitment or product sale (31.205-1), and excessive executive compensation above the statutory cap. When DCAA finds unallowable costs in indirect pools, the government is owed a credit adjustment.

A contractor's wrap rate multiplier is 3.8× their direct labor rate. Compared to an industry average of 3.0×, what should a PM do?

- A. Accept it since contractors set their own rates

✓ B. Request an explanation and FPRA documentation; question overhead and G&A rates specifically

- C. Immediately terminate the contract
- D. Use the rate only if the contract is cost-plus

WHY

A 3.8× multiplier is above typical industry range (2.5-3.5× for most defense contractors). The PM should request the FPRA letter or rate documentation, identify which pool (overhead or G&A) is driving the high rate, and challenge rates that are not supported by audited actuals. High rates may indicate excessive overhead, unallowable costs in pools, or the contractor's rates not being DCAA-audited.

What is the purpose of a contractor's annual Incurred Cost Submission (ICS)?

A. To propose new labor categories for the upcoming year

✓ **B. To report actual indirect cost rates for the completed year, enabling reconciliation with billed rates**

C. To submit a new FPRA request

D. To report earned value performance data

WHY

The ICS (also called Incurred Cost Proposal) reports a contractor's actual indirect rates for the fiscal year, which are compared to the provisional or FPRA rates that were billed to contracts during the year. If actual rates were lower than billed, the contractor owes a credit. If higher, the government may owe additional costs on cost-reimbursable contracts. DCAA audits ICS submissions: delays of years are common and represent a significant financial risk to the government.

The Weighted Guidelines Method (DFARS 215.404-71) is used to determine:

A. The period of availability of a specific appropriation type

✓ **B. A target profit/fee rate on negotiated defense contracts based on risk, contract type, and capital employed**

C. The contractor's overhead rate pool structure

D. The maximum labor categories allowed in a proposal

WHY

The Weighted Guidelines Method scores five factors (performance risk, contract type risk, facilities capital employed, cost efficiency, and independent R&D) to produce a target profit percentage. It ensures profit is commensurate with the risk and investment a contractor bears: higher risk contracts earn higher fee rates. Government negotiators use this analysis to ensure profit is fair but not excessive.

Fringe benefits are applied to which cost base in a standard defense contractor cost buildup?

A. Total Contract Value

B. All indirect costs

✓ **C. Direct Labor Rate: fringe is a percentage of the employee's direct labor**

D. G&A base only

WHY

Fringe benefits (FICA, health insurance, retirement contributions, vacation/sick leave accrual, workers' comp) are applied as a percentage of direct labor dollars. They represent the employer's cost of employee benefits above base salary. A fringe rate of 32% on a \$70/hr DLR adds \$22.40/hr, representing the employer's share of the employee's total compensation package.

DEFENSE FINANCE & BUDGETING · LESSON 9

DCAA & DCMA: Your Two Most Important Oversight Partners

20 min

Understand the roles, authorities, and tools of DCAA and DCMA, and how to work with them effectively to protect your program and the taxpayer.

The Auditors and the Administrators

Most people in defense contracting hear DCAA or DCMA and feel a mild sense of dread. Wrong reaction. DCAA (Defense Contract Audit Agency) looks at the money: are the costs you're claiming allowable, allocable, and reasonable? DCMA (Defense Contract Management Agency) looks at the work: are you delivering what the contract says, and is your program data reliable? Both agencies carry “Defense” in the name, so if the rebrand to Department of War ever reaches the acronym level, brace for “WCAA” and “WCMA” showing up in someone's slide deck with a straight face. Think of DCAA as the financial auditor and DCMA as the performance monitor. If you're keeping accurate records, spending on legitimate costs, and delivering what you promised, these agencies are your allies, not your adversaries.

Two Agencies, Two Missions: Both Critical to Your Program

Every major defense PM operates within a three-way oversight structure: the PCO (Procuring Contracting Officer) awarded the contract, DCMA administers it day-to-day, and DCAA audits the financial side. Many new PMs confuse DCAA and DCMA: they have different missions, different authorities, and different relationships with your program. Understanding what each can and cannot do determines whether you use them as effective oversight tools or accidentally create friction that slows your program down.

DCAA vs. DCMA: Side-by-Side Comparison

ATTRIBUTE	DCAA	DCMA
Full Name	Defense Contract Audit Agency	Defense Contract Management Agency
Mission	Financial auditing (audit contractor accounting systems, cost proposals, and incurred costs)	Contract administration) oversee contractor performance, quality, and schedule at contractor facilities
Established	1965 (Secretary of Defense directive)	2000 (formerly Defense Contract Management Command)
Workforce	~4,900 civilian auditors	~12,500 civilian and military personnel
Key Output	Audit reports (favorable/qualified/adverse) to ACO/PCO	Contractor Performance Assessment Reports (CPARs), EVMS validation, consent to subcontract decisions
Authority Over Contracts	None (DCAA advises; ACO/PCO decide)	Yes) ACOs have direct contract administration authority
Physical Location	At DCAA field offices (resident or visiting teams)	At contractor facilities (resident teams for major contractors)
Relationship to PM	Independent (PM cannot direct DCAA work)	Collaborative) PM works with ACO daily on contract administration

DCAA: What They Actually Audit

DCAA does not audit every contractor: it prioritizes its limited resources on the highest-risk contracts. The major audit types every PM should understand: (1) Pre-Award Surveys: before award of a cost-reimbursable contract, DCAA may audit the contractor's accounting system to confirm it meets FAR requirements. An “inadequate” accounting system can block contract award. (2) Incurred Cost Audits: after each fiscal year, DCAA audits the contractor's ICS to verify actual indirect rates and ensure only allowable costs were billed. These audits often take years to complete, creating significant cost uncertainty. (3) Forward Pricing Rate Audits (DCAA audits proposed indirect rates to support FPRA negotiations. (4) Truth in Negotiations Act (TINA) / Defective Pricing Audits) post-award audits to verify that cost or pricing data submitted before award was accurate, current, and complete.

The Accounting System Adequacy Finding: A PM's Red Flag

When DCAA finds a contractor's accounting system “inadequate,” it is a program emergency. An inadequate finding means the system does not properly identify, accumulate, and segregate costs by contract, which means you cannot trust the costs being billed to your program. In extreme cases, the ACO may suspend contractor billing or require withholds (typically 5-10% of billings) until the system is corrected. If your contractor has an open inadequate accounting system finding, flag it to your PCO immediately and track the remediation plan.

DCMA: Your Government Representative at the Contractor's Facility

DCMA is the operational arm of government contract oversight. On major programs, DCMA assigns a resident Administrative Contracting Officer (ACO) and team directly to the contractor's facility. This team monitors schedule, quality, property management, and contract compliance on a daily basis. DCMA's ACO has the authority to: consent to or disapprove major subcontracts, accept or reject deliverables, issue cure notices, issue show cause letters before termination, and manage Government-Furnished Property (GFP). For PMs, the ACO is your proxy at the contractor: your most direct source of real-time performance intelligence.

DCMA Key Functions and PM Interactions

DCMA FUNCTION	WHAT THEY DO	PM'S ROLE
EVMS Validation	Validates contractor EVMS meets ANSI/EIA-748 before government accepts it for reporting	PM coordinates with DCMA; without validation, IPMR data is not credible
Contract Administration	ACO manages billing, modifications, property, subcontracts day-to-day	PM works with ACO on mod approvals, billing disputes, and performance issues
Quality Assurance	QARs witness testing, inspect deliverables, maintain Product Acceptance	PM ensures QAR access; ACO signs final acceptance on deliverables
Consent to Subcontract	ACO approves major subcontracts when required by contract clause	PM ensures subcontract consent is included in program schedule: delays cost time
Cure Notice / Show Cause	ACO issues cure notice when contractor is at risk of default; show cause before termination	PM must be informed: these are escalation steps toward potential termination
CPARS Entries	DCMA contributes to CPARS assessments on contractor performance	PM provides input; CPARS records follow contractor for 3 years in future source selections

DFARS Business Systems: The Six Systems DCMA and DCAA Watch

DFARS 252.242-7005 (Contractor Business Systems) identifies six contractor business systems that are subject to government review and withholding if found “significant deficiencies”: (1) Accounting System (DCAA audits), (2) Estimating System (DCAA audits), (3) Earned Value Management System (EVMS (DCMA validates), (4) Purchasing System (DCMA reviews), (5) Material Management and Accounting System) MMAS (DCMA), (6) Property Management System (DCMA). When DCAA or DCMA finds a “significant deficiency” in any of these systems, the ACO can withhold 5% of contract billings (up to 10% total). As a PM, a withheld billing is cash flow risk for your contractor and a potential harbinger of deeper systemic problems.

Do Not Obstruct DCAA Access

DCAA auditors have the right to access contractor records, employees, and facilities under FAR 52.215-2 (Audit and Records: Negotiation). A contractor that limits, delays, or obstructs DCAA access is committing a serious FAR violation. Equally important: as a government PM, never pressure a contractor to limit DCAA access to manage a schedule or cost outcome. This constitutes improper interference with an audit: a federal law violation with serious career and legal consequences. If a contractor claims they cannot provide records to DCAA, contact your PCO and legal office immediately.

Working With Your ACO: Practical Tips

The most effective PMs build a strong relationship with their DCMA ACO. Best practices: (1) Include the ACO in your monthly program reviews, they are your government ally at the contractor, not the enemy combatant your gut may initially suggest. (2) Read every DCMA report your ACO produces, they often contain early warnings before IPMR data shows problems. (3) Ensure your contract includes appropriate DCMA access clauses (FAR 52.246-2, 52.242-4). (4) Do not go around your ACO to address contractor performance issues, work through them. (5) For CPARS, get the ACO's performance input before finalizing your assessment, their daily visibility is invaluable.

Working the DCAA/DCMA Relationship to Your Advantage

Mid-career PMs who treat DCAA and DCMA as adversaries are leaving oversight resources on the table. Both agencies work for the government: their findings protect your program, not the contractor. Establish a quarterly touchpoint with your ACO and DCAA auditor: share your program priorities, the highest-risk WBS elements, and any contractor behaviors that concern you. DCAA can target its audit resources to your hot spots if you tell them where to look. Request a floor check audit when your contractor reports a staffing surge that isn't reflected in performance. Ask DCMA to review the contractor's subcontract management practices when you see a subcontractor performance problem the prime isn't addressing. These agencies have authorities you don't: use them.

FROM LUCAS

Senior PMs have mechanisms to financially pressure non-performing contractors that junior PMs rarely use. For cost-type contracts, DCAA can recommend disapproval of incurred cost submissions: triggering a withhold of up to 5% of billed costs pending resolution. For fixed-price contracts with performance milestones, you can withhold milestone payments when deliverables fail to meet contract requirements. The key: document the basis for the withhold in writing, citing specific contract clauses (FAR 52.232-7 for cost; the specific milestone clause for FP). Contractors will push back aggressively and threaten REAs. That's acceptable. What's not acceptable: using withholds as negotiating leverage for contract modifications (that's coercion). Withholds must be based on legitimate contract performance deficiencies, not as a tool to extract concessions.

Key Terms

DCAA

Defense Contract Audit Agency: the DoD agency responsible for auditing contractor accounting systems, cost proposals, and incurred cost submissions. Provides independent financial advice to contracting officers.

DCMA

Defense Contract Management Agency: the DoD agency responsible for contract administration, quality assurance, and contractor performance oversight on major contracts. DCMA is the government's "eyes and ears" at contractor facilities.

ACO

Administrative Contracting Officer: a DCMA contracting officer delegated authority to administer specific aspects of a contract (billing, consent to subcontract, property management).

QAR

Quality Assurance Representative: a DCMA official who monitors contractor quality processes and product acceptance at the contractor's facility.

EVMS Compliance

DCMA validates that contractor EVMS meets ANSI/EIA-748 criteria before government accepts it for IPMR reporting. A non-compliant EVMS is a major program risk.

Accounting System Adequacy

DCAA determination that a contractor's accounting system properly identifies, accumulates, and reports costs in accordance with FAR Part 31 and CAS. Required before award of cost-reimbursable contracts.

Billing System

The contractor's system for submitting invoices to the government. DCAA audits billing systems for accuracy and compliance.

Estimating System

The contractor's system for preparing cost proposals. DCAA audits estimating systems to ensure proposed costs are reasonable and consistent with actuals.

CIPR

Contractor Insurance and Pension Review: DCAA audit verifying that insurance and pension costs charged to government contracts are allowable and allocable.

Provisional Billing Rates

Interim indirect rates approved by the ACO/DCAA for billing purposes while actual rates are being finalized in the ICS process.

Knowledge Check

What is the primary mission of DCAA?

A. Administer contracts and oversee contractor performance at contractor facilities

✓ **B. Provide independent financial auditing of contractor accounting systems and cost proposals**

C. Award defense contracts and negotiate contract modifications

D. Certify contractor EVMS compliance with ANSI/EIA-748

WHY

DCAA's mission is financial auditing: it audits contractor accounting systems, cost proposals, and incurred cost submissions to protect the government's financial interests. DCAA does not administer contracts (that is DCMA's ACO) and does not have contract award authority (that is the PCO). EVMS certification is DCMA's function.

An Administrative Contracting Officer (ACO) from DCMA has the authority to:

A. Award new contracts independently

✓ **B. Withhold contractor billings for EVMS deficiencies, consent to subcontracts, and accept deliverables**

C. Conduct financial audits of contractor cost proposals

D. Authorize Nunn-McCurdy breach notifications

WHY

The ACO has direct contract administration authority: including consent to subcontract decisions, GFP management, billing approval, cure notices, and deliverable acceptance. The ACO can withhold up to 5% of billings (10% maximum total) when contractor business systems have significant deficiencies. DCAA conducts audits (separate from ACO), and PCOs award contracts.

A DCAA “inadequate” accounting system finding on your prime contractor most immediately means:

A. The contract must be terminated for default

✓ B. Cost billings may not be trustworthy; the ACO may withhold a portion of billings until the system is corrected

C. The contractor must rebid the contract

D. DCMA takes over all contractor financial management

WHY

An inadequate accounting system finding means DCAA cannot rely on the contractor's cost data, which means billed costs may not accurately reflect what was charged to your contract. The ACO typically implements a billing withhold (5% to 10%) pending remediation. The PM should elevate this to the PCO and monitor the corrective action plan closely.

Which of the six DFARS Business Systems does DCMA validate (not DCAA)?

A. Accounting System

B. Estimating System

✓ C. Earned Value Management System (EVMS)

D. Billing System

WHY

DCMA is responsible for EVMS validation: confirming the contractor's EVM system meets ANSI/EIA-748 criteria. DCAA audits the Accounting System, Estimating System, and Billing System. The split reflects each agency's expertise: DCMA understands program performance systems; DCAA understands financial accounting systems.

Under DFARS 252.242-7005, when DCAA or DCMA finds a “significant deficiency” in a contractor business system, the maximum billing withhold is:

A. 2%

B. 5%

✓ C. 10%

D. 25%

WHY

DFARS 252.242-7005 allows withholds of 5% per deficient system, up to a maximum of 10% total across all systems. These withholds are intended to incentivize rapid system correction while preserving contractor cash flow to the extent possible. Withholds are lifted when the ACO approves a corrective action plan or confirms the deficiency has been resolved.

An Incurred Cost Submission (ICS) is filed by contractors to:

A. Propose rates for the upcoming fiscal year

✓ **B. Report actual indirect cost rates for a completed fiscal year, enabling reconciliation with billed rates**

C. Submit progress payments for completed milestones

D. Certify EVMS compliance to DCMA

WHY

The ICS (filed within 6 months after fiscal year end) reports actual indirect rates for the year. DCAA audits these against the provisional rates billed to government contracts. If the contractor over-billed (provisional rates higher than actuals), the government gets a credit. If under-billed, the government owes additional cost on cost-reimbursable contracts. ICS audit backlogs of 5-10 years create significant financial uncertainty.

A DCMA Quality Assurance Representative (QAR) at a contractor facility primarily performs:

A. Financial audits of contractor billing records

✓ **B. Oversight of contractor quality processes, product inspections, and delivery acceptance at contractor facilities**

C. Negotiation of contract modifications

D. Performance of independent cost estimates

WHY

QARs are DCMA's quality oversight representatives resident at contractor facilities. They witness testing, inspect deliverables, monitor quality management systems, and sign off on acceptance. Their daily presence gives the government real-time visibility into product quality: long before a PM receives formal reports or delivery.

The Truth in Negotiations Act (TINA) / Defective Pricing audit by DCAA is triggered when:

A. Any contract over \$350K is awarded

✓ **B. A contract over \$2M required the submission of certified cost or pricing data before award**

C. A contractor files an ICS late

D. A Nunn-McCurdy breach is declared

WHY

TINA (codified as 41 U.S.C. § 3502) requires contractors to submit certified cost or pricing data for contracts exceeding \$2M (where competition and other exceptions do not apply). Post-award, DCAA may audit whether that data was accurate, current, and complete at the time of submission. If data was defective (incorrect or outdated), the government is entitled to a price reduction equal to the amount overpaid.

Which statement about DCAA access to contractor records is correct?

A. Contractors may limit DCAA access to protect proprietary information

✓ **B. DCAA has the right under FAR 52.215-2 to access all relevant records; obstructing this access is a FAR violation**

C. DCAA access requires a court order for private contractor records

D. DCAA can only access records voluntarily provided by the contractor

WHY

FAR clause 52.215-2 (Audit and Records, Negotiation), included in negotiated contracts, gives the government (and DCAA) the right to examine and audit all records related to the contract. Contractor obstruction of DCAA access is a serious FAR violation. PMs must ensure contractors understand their audit access obligations and escalate access disputes immediately to the PCO and legal.

In the context of DFARS Business Systems, a “Purchasing System” review by DCMA focuses on:

A. The contractor's system for purchasing commercial supplies

✓ **B. The contractor's system for acquiring subcontracted work and materials: ensuring competition, cost analysis, and flowdown of contract requirements**

C. The contractor's enterprise resource planning (ERP) system

D. The contractor's EVMS purchasing module

WHY

DCMA's Purchasing System review examines how the contractor manages its supply chain: whether it properly competed subcontracts, performed cost/price analysis on non-competitive subs, and flowed down required government contract clauses to subcontractors. Poor purchasing system controls can result in overbilled subcontract costs, undisclosed subcontractor teaming, and failure to meet small business subcontracting goals.

DEFENSE FINANCE & BUDGETING · LESSON 10

Contract Finance Intelligence: CPAF, Burn Rate & Award Fee

55 min

Why how you describe your work determines how much your company is worth, and how shifting from logistics to engineering changes your financial picture.

Why This Lesson Exists

Most PMs never think about how revenue gets reported. The way your company describes your work on paper determines how much the business is worth. Two companies can do identical work on the same DoD contract and end up with wildly different financial profiles depending on whether they are classified as a Principal or an Agent, the accounting equivalent of two people doing the same job with very different titles on their business cards.

The 5-Step Revenue Test

To recognize revenue as a Principal under ASC 606, your company must satisfy five steps. As a PM, your job is to make sure the work you manage is documented in a way that passes each step.

FROM LUCAS

I've sat in rooms where a company was pitching to a potential acquirer and the first question was "are you booking gross or net?" If the answer is net, the valuation conversation changes immediately. Your CPARS might be perfect, your team might be excellent, but on paper you're a pass-through. That matters when you're trying to grow, borrow money, win larger contracts, or sell the company.

The Hardware Dump Problem

One of the most common traps in defense contracting is the hardware dump: a Statement of Work that only asks for hardware delivery with no engineering work described. If your SOW just says "deliver 50 servers," you're an Agent. But if that same SOW says "deliver, configure, harden, integrate, and test 50 servers per the government's cybersecurity requirements," you're a Principal. Same servers, same box truck, very different balance sheet.

If It Is Not Documented, It Did Not Happen

From an auditor's perspective, verbal work and undocumented engineering are invisible. If your team spent 300 hours integrating a system but the SOW does not describe integration as a performance obligation, that work does not support a Principal classification.

Key Terms

CPAF

Cost-Plus Award Fee: a contract type where the government reimburses 100% of allowable costs and pays two layers of fee: a small fixed base fee (max 3%) and a variable award fee earned through performance ratings.

Award Fee

The variable, performance-linked portion of contract profit. Earned only for above-satisfactory performance across rated factors (Technical, Cost, Schedule, Management). Unsatisfactory performance earns \$0 regardless of other factors.

Base Fee

The fixed, guaranteed floor of profit on a CPAF contract. Capped at 3% of estimated contract cost under DFARS 216.405-2. Paid regardless of performance scores.

Wrap Rate

The cost multiplier applied to every dollar of direct labor: $\text{Billed Cost} = \text{Base Salary} \times (1 + \text{Fringe}) \times (1 + \text{Overhead}) \times (1 + \text{G\&A})$. A single employee at \$100K base salary can bill \$200K+ to the government after wrap rates are applied.

Burn Rate

The actual monthly cost spend rate on a contract. Finance calculates it weekly. If burn rate exceeds the remaining funded value, all work must stop: spending beyond obligated funding violates the Anti-Deficiency Act (ADA), a federal offense.

Labor Utilization

$\text{Direct Hours} \div \text{Total Available Hours}$. The percentage of your team's time charged to direct (billable) work. Target is 75–85%. Below 70% inflates overhead rates company-wide; above 90% risks burnout and quality degradation.

Anti-Deficiency Act (ADA)

Federal law prohibiting government contractors from spending beyond officially obligated funding. An ADA violation is a federal criminal offense. The funded ceiling is set by contract modification and ACRN: never exceed it.

EAC (Estimate at Completion)

Finance's projection of total contract cost by end of performance, based on current burn rate and remaining scope. Used to detect funding gaps before they become crises. PMs should sync with Finance on EAC weekly.

ODCs (Other Direct Costs)

Non-labor direct costs charged to a contract: travel, equipment, materials, training. High-risk cost area: require pre-approval and documented contract nexus. Non-mission travel and unrelated training are waste that drains award fee.

DCAA

Defense Contract Audit Agency. Audits contractor indirect rates (fringe, overhead, G&A) and cost pools to ensure compliance with FAR Part 31. Unallowable costs (entertainment, alcohol, image advertising) must be excluded from all pools.

Knowledge Check

On a CPAF contract, what happens to the company's profit if you save costs by finishing work under budget?

- A. Profit increases dollar-for-dollar because costs saved become profit
- ✓ B. Profit only improves if the cost savings improve your award fee score
- C. Profit decreases because you billed less to the government
- D. There is no impact on profit because the government pays a fixed amount

WHY

CPAF is a cost-reimbursement vehicle; the government pays your costs either way. Profit comes from base fee (fixed, small) and award fee (variable, performance-driven). Reducing costs only improves profit indirectly, by boosting your cost control evaluation score and thus your award fee earned.

The government scores your award fee across four factors. Cost Control must represent at least what percentage of the total evaluation weight?

- A. 10%
- B. 15%
- ✓ C. 25%
- D. 35%

WHY

Cost Control must represent at least 25% of the evaluation weight per federal acquisition regulations. In a typical CPAF structure: Technical 35%, Cost Control 25%, Schedule/Timeliness 25%, Management/Customer 15%. Every PM decision that affects cost is directly affecting at least one quarter of the total award fee.

Your burn rate is running significantly below plan for the past two months. Which of the following is NOT a likely consequence?

A. The government may de-obligate unused funds at fiscal year end

B. Your schedule evaluation score may suffer

✓ C. Your award fee will automatically increase because you spent less

D. Finance may flag staffing gaps or scope issues

WHY

Under-execution (burn rate too low) is a problem, not a reward. It risks fund de-obligation, damages schedule scores, and signals delivery problems to the government. Award fee isn't earned by spending less: it's earned by performing well across all four evaluation factors.

What is the maximum base fee allowed on a CPAF contract?

A. 1% of estimated contract cost

✓ B. 3% of estimated contract cost

C. 10% of estimated contract cost

D. There is no cap: it is negotiated freely

WHY

DFARS 216.405-2 limits base fee to 3% of estimated contract cost. The base fee is intentionally small: it's the floor, not the target. The real profit opportunity is in the award fee pool, which incentivizes above-satisfactory performance. PMs who maximize performance scores earn multiples of what the base fee alone provides.

A subcontractor on your contract has been performing tasks not described in their Statement of Work (SOW) because it was convenient. What is the primary risk?

✓ A. The subcontractor may bill for the extra work, creating unallowable or disputed costs

B. The subcontractor will receive a lower performance rating

C. The contract will be terminated for convenience

D. No risk: if the work benefited the government, it will be reimbursed

WHY

Subcontractor scope discipline is a financial control. Work performed outside an approved SOW lacks proper authorization under FAR Part 31 cost principles. An audit can disallow those costs, meaning the company absorbs them with no government reimbursement. Beyond the direct cost risk, loose SOW management also signals poor oversight to the government evaluator: damaging both cost control and management scores.